



V93000 Specifications

V93000 SOC Pin Scale 9G Digital Card

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PS9G specifications

The Pin Scale 9G (PS9G) is a high-speed digital card for the V93000 ATE platform.

Note

The specifications are preliminary and are subject to change.



Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

PS9G overview

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, *valid system calibration* and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Test Processor Per-Pin architecture

Maximum channel count	64 channels per card
Parallel vector memory ^[1]	16 MByte (64 MVectors at x4 mode) standard, 32 MByte (128 MVectors at x4 mode) optional, 64 MByte (256 MVectors at x4 mode) optional, 112 MByte (448 MVectors at x4 mode) optional
Maximum scan depth ^[2]	28.5 GVectors (at x4mode)

Speed

Maximum data rate	> 8000 Mbps (x8 mode) 6800 Mbps (x6 mode)
Maximum clock rate	> 4 GHz

Maximum external input voltage (compliance range)

Maximum range for external voltage applied to pin ^[3]	-6.0V to +9.0V
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Test programming and debug

The SmarTest production test environment includes manufacturing test interfaces to operators, handling equipment, and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test methods. User-specific tests are programmed with test methods in C++. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots, pin margin tests and memory bit map.

Compatibility

PS9G is supported starting with SmarTest 7.1 and higher. It can be used with with existing Pin Scale, DC Scale and Wave Scale RF cards. .

^[1] vector memory is used for pattern, setup data and result capture

^[2] Maximum scan depth support total of 8 chains per instrument, 3.584GVector max (drive only, single bit mode) or 1.792 GVectors in x4 mode for EACH chain. Total data volume per instrument ~ 28.5 GVectors (in x4 mode).

^[3] valid for DC connection; for AC connection -1V to 4V (50 Ohm termination)

Calibration and operation

Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months
Calibration period ^[4] (system auto adjustment)	3 months

Environmental

Operating	15°C to 30°C (59°F to 86°F)
Specification warranted temperature	20°C to 30°C (68°F to 86°F)
Maximum humidity at 30°C	< 70% R. H., non-condensing
Storage (without water)	-40°C to +70°C

^[4] valid at ambient temperature within ± 2.5 K of calibration temperature. For temperature requirements during calibration please see "Maintenance Guide."

PS9G Timing

PS9G OTA/EPA

For OTA/EPA test method condition: TX to RX loop using PRBS 2^{15} pattern and verified with Vswing = 250mV and 850mV and speeds up to 6.8 Gbps single ended mode and up to 8.0 Gbps differential mode, loop includes standard high speed FR4 DUT board traces

OTA/EPA Single Ended

	4.5 Gbps per pin pair	4.5 Gbps board 64 pins	6.8 Gbps per pin pair	6.8 Gbps board 64 pins
Overall Timing Accuracy (t_{OTA})	< ± 35 ps	< ± 46 ps	< ± 44 ps	< ± 55 ps
Edge Placement Accuracy ($t_{EPA}=t_{OTA}/2$)	< ± 18 ps	< ± 23 ps	< ± 22 ps	< ± 28 ps

OTA/EPA Differential

	4.5 Gbps per lane	4.5 Gbps board 16 lanes	8 Gbps per lane	8 Gbps board 16 lanes
Overall Timing Accuracy (t_{OTA})	< ± 30 ps	< ± 38 ps	< ± 40 ps	< ± 55 ps
Edge Placement Accuracy ($t_{EPA}=t_{OTA}/2$)	< ± 15 ps	< ± 19 ps	< ± 20 ps	< ± 28 ps

Edge Placement

Edge placement resolution	1 ps
Edge placement range ^[5]	-4 to 32 Sequencer periods

PS9G Vector Period

Minimum vector period ^[6]	125 ps
Vector Period Resolution	0.001 fs
Vector Period Accuracy	± 15 ppm of period setting
Minimum sequencer period	880 ps
Maximum sequencer period	5000 ns

^[5] edge placement range maximal -5000 ns to +15000 ns

^[6] x8 mode for differential drive and receive; over-programming possible

PS9G Eye Height

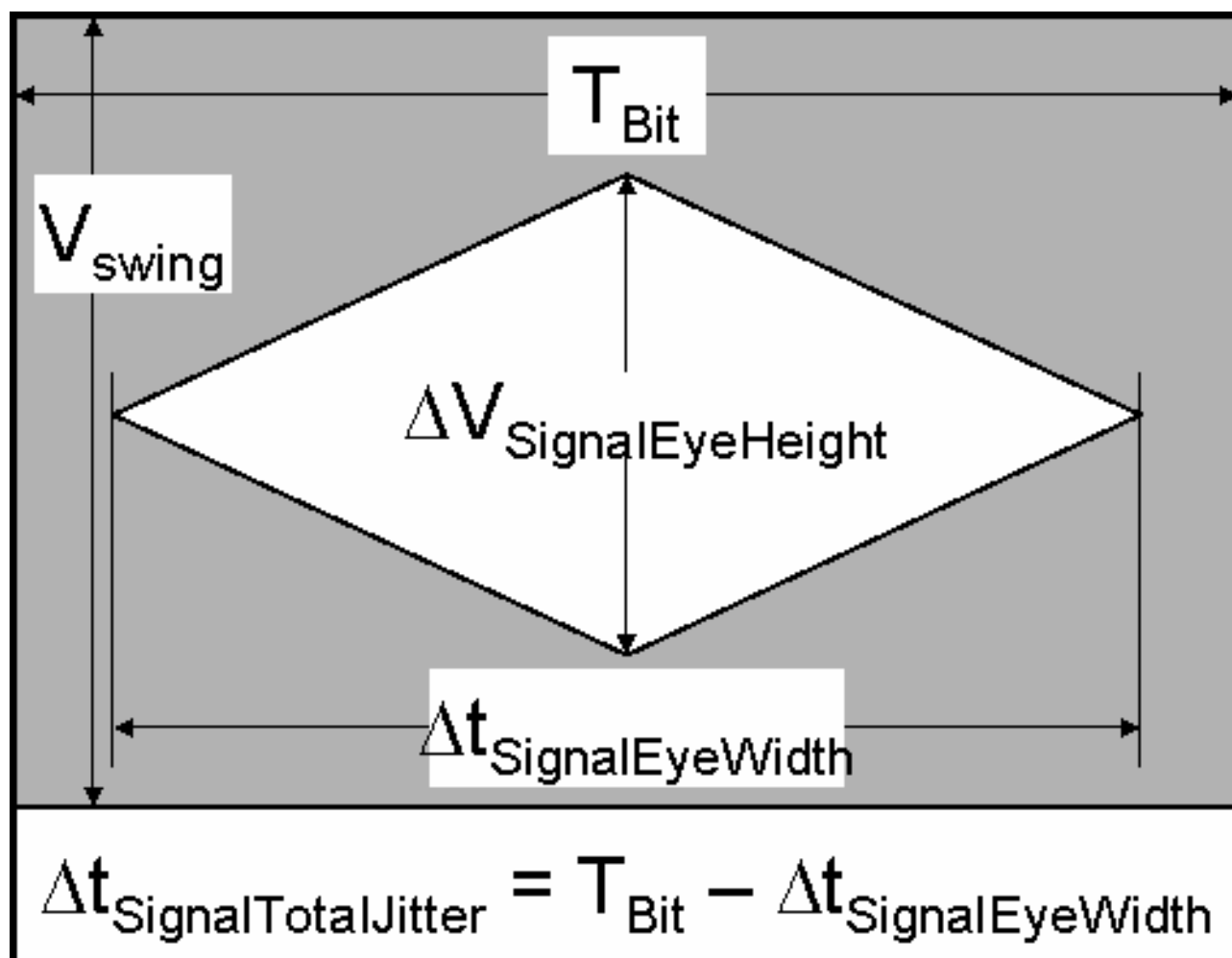
Single Ended (verified with $V_{swing} = 850\text{mV}$ into 50Ω)

	2.5 Gbps	5.33 Gbps	6.8 Gbps
Typical $\Delta V_{\text{SignalEyeHeight}}$ of driver	89 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{\text{SignalEyeHeight}}$ of driver+receiver pair	88 % of V_{swing}	68 % of V_{swing}	54 % of V_{swing}
Minimum $\Delta V_{\text{SignalEyeHeight}}$ of driver+receiver pair ^[7]	78 % of V_{swing}	29 % of V_{swing}	15 % of V_{swing}

Differential

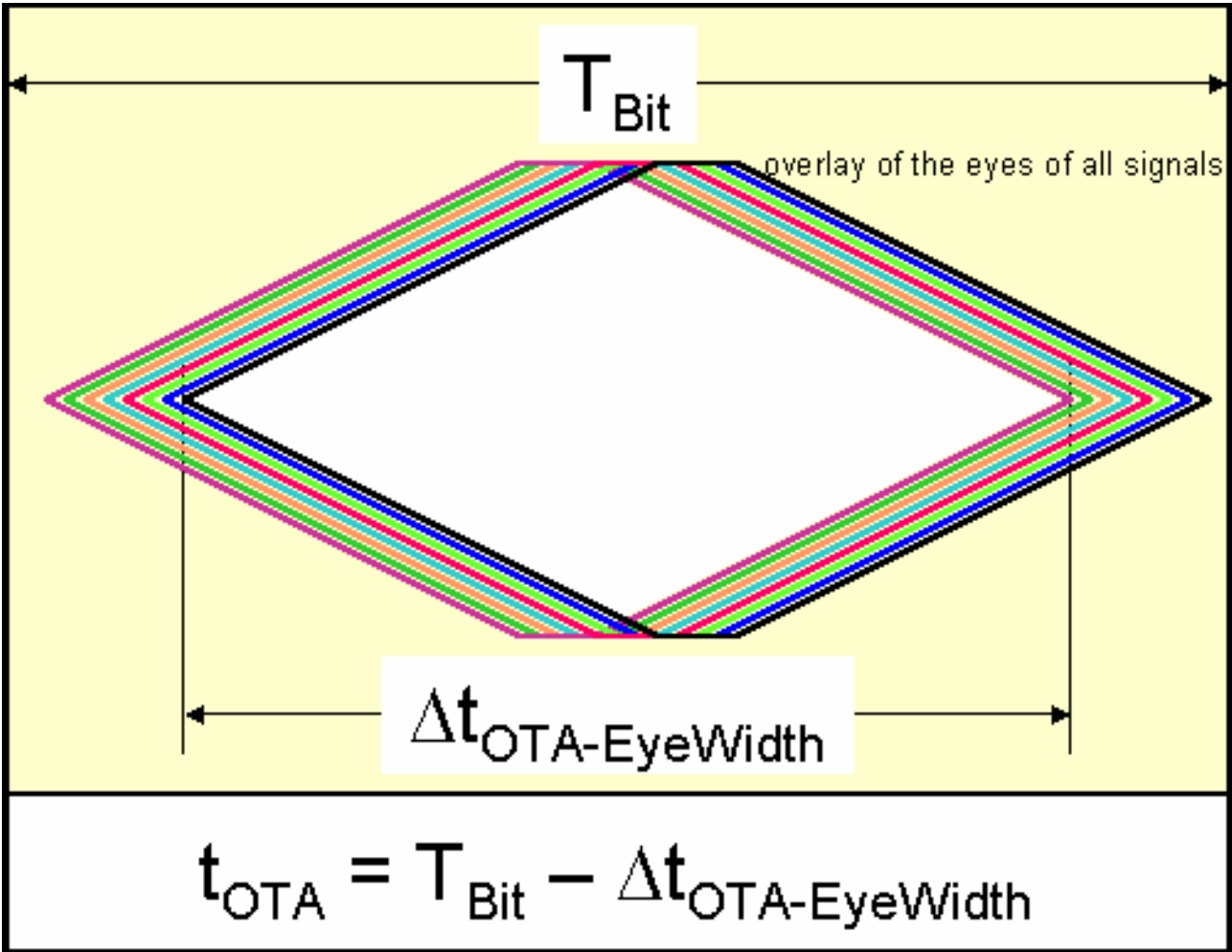
	2.5 Gbps	5.33 Gbps	8.0 Gbps
Typical $\Delta V_{\text{SignalEyeHeight}}$ of driver	85 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{\text{SignalEyeHeight}}$ of driver+receiver pair	80 % of V_{swing}	68 % of V_{swing}	35 % of V_{swing}
Minimum $\Delta V_{\text{SignalEyeHeight}}$ of driver+receiver pair	74 % of V_{swing}	37 % of V_{swing}	11 % of V_{swing}
Minimum $\Delta V_{\text{SignalEyeHeight}}$ of driver+receiver pair			

^[7] tester driver measured with tester receiver using PRBS 2^{15} pattern, "equalization (Topic 343254)" is active.

OTA Specification

Total Jitter ($\Delta t_{\text{SignalTotalJitter}}$)

Eye Diagram



Eye Diagram

PS9G Linearity

The graph below describes the non-linearities of the edge delays between the driver (programmed delay) and the receiver (measured delay) even after calibration.

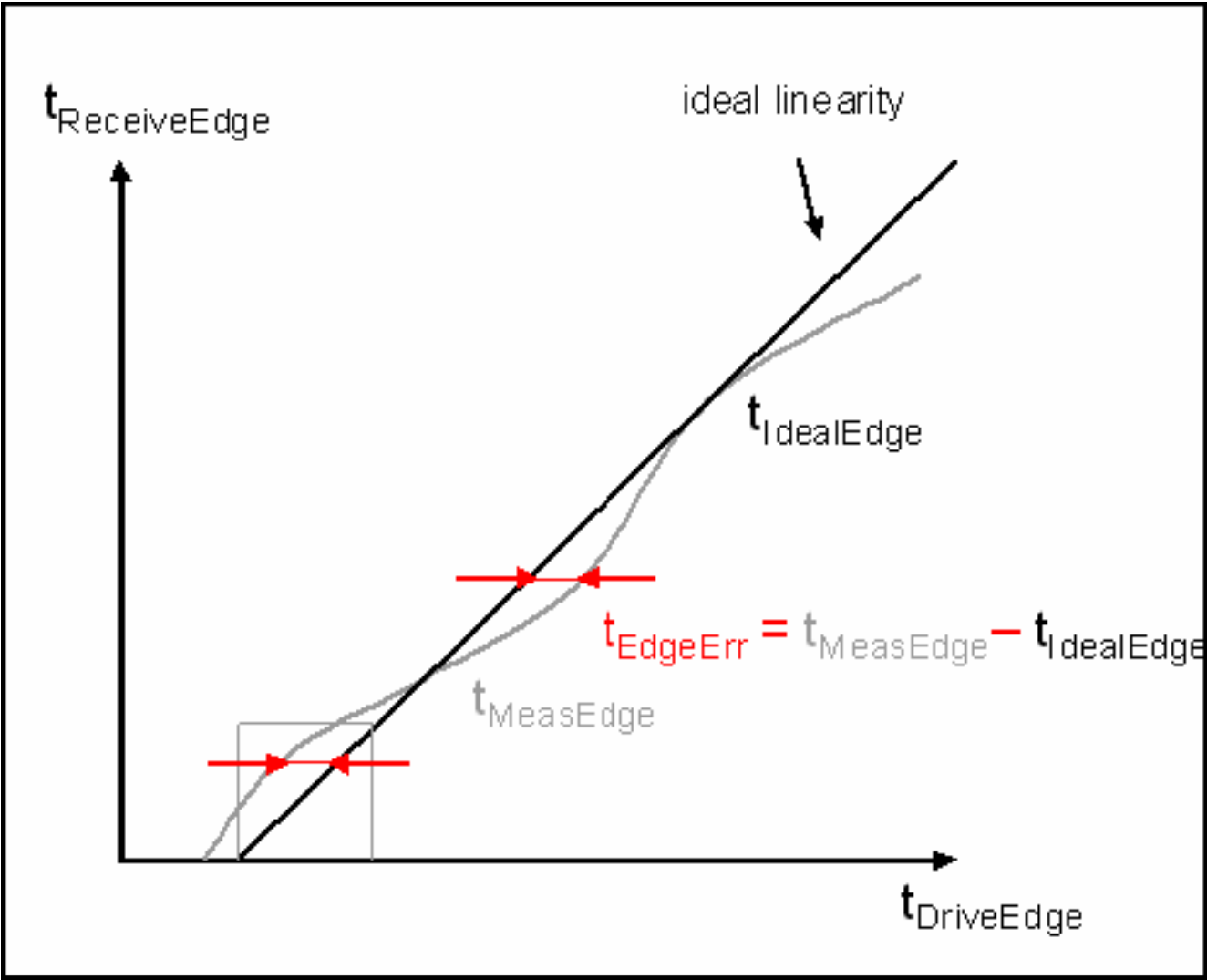
The black line represents the ideal delay, that is, the receiver delay follows the driver delay and may have an offset. The gray line represents a typical measurement. Here, the measured delay differs from the ideal delay by an edge delay offset (t_{EdgeErr}). This offset could be either a differential non-linearity (point error) or an integral non-linearity (integral error over a delay range).

The driver and receiver edges are treated independently and contribute equally to the uncertainty of the edge placement. Therefore, the edge placement non-linearity would be double of what is specified here.

Note These non-linearity terms are part of the PS9G OTA/EPA specifications and do not need to be considered separately.

Integral Non-Linearity ^[8] INL ($ t_{\text{EdgeErr}} $)	±5 ps
Differential Non-Linearity DNL	5 ps

^[8] Single ended driver measured with single ended receiver and differential drive measured with differential receiver.



PS9G Clocks

Differential clock^[9] jitter at 4.0 GHz

typical 2.9 ps rms

PS9G Clock Data Recovery (CDR)

Tracking ^[10] range	±7.5 UI
Tracking max frequency without clock forwarding	100 MHz
Tracking max frequency with clock forwarding	1 MHz

^[9] Tester driver measured with tester receiver and measured value divided by $\sqrt{2}$. Swing is in range 250mV..850mV

^[10] Tracking operates in x4 and x6 mode, max speed limit is depending on high frequency jitter amplitude of device, this means typical speed limit is in range 3..4Gbps.

PS9G Drive Jitter Injection

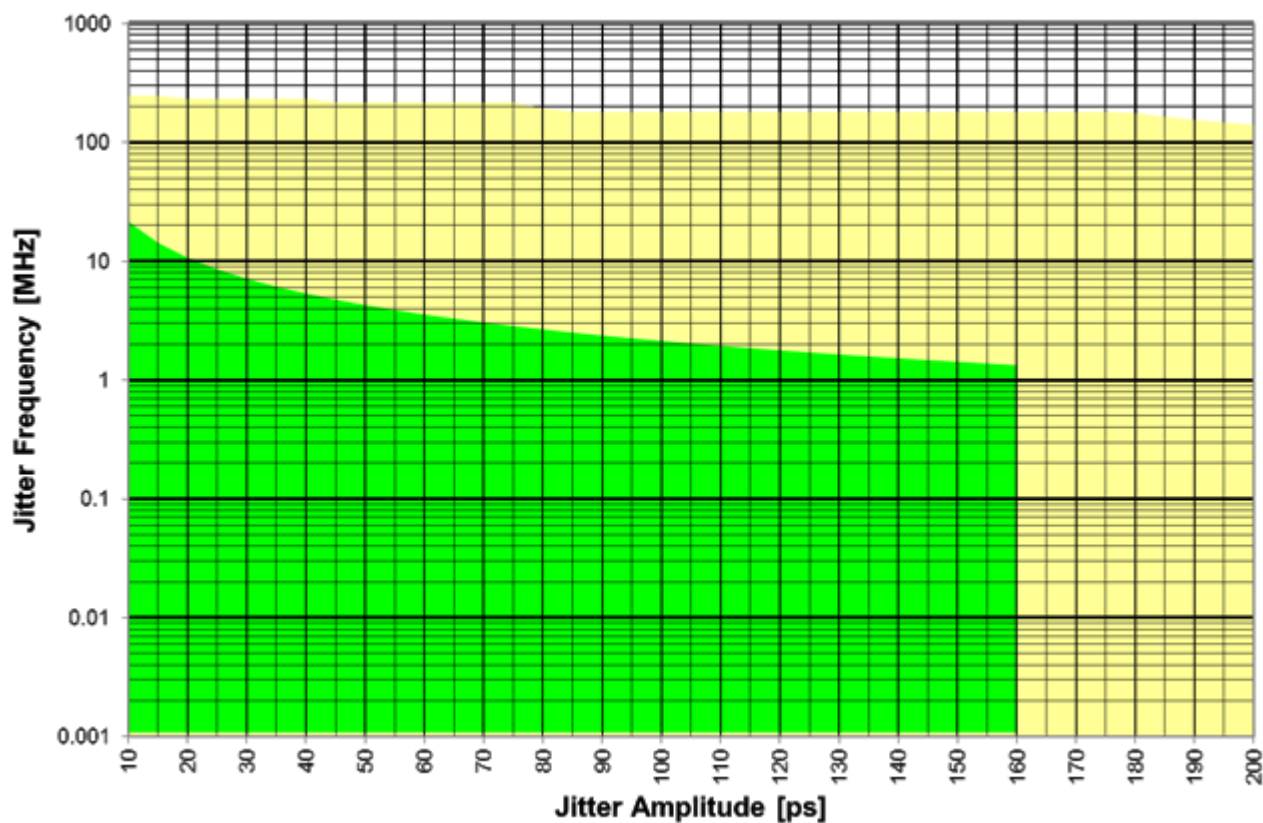
Jitter Injection via delay modulation with digital per pin AWG with 512 samples.

Timing Jitter range^[11]

400 ps Peak To Peak

Jitter Frequency range^[11]

1 kHz to 110 MHz



Drive Jitter Injection

^[11] Jitter Injection Delay Modulation AWG sampling rate depends on active timing. Sample value range is -200ps..+200ps, some limitations do apply.

PS9G Board ADC

The Board ADC provides very accurate high impedance voltage measurements behind each Pin Scale 9G channel. A Board ADC is shared among 16 channels.

Voltage range	-5.0 V to 8.0 V
Voltage resolution	300 μ V
Accuracy	$\pm$ 2 mV
Input impedance	> 10 MOhm

PS9G Driver

AC Performance

Transition time at 850 mV swing in 50 Ω (20 to 80%) ^[12] 65 ps $\pm$ 15 ps

DC Performance (specified into open and maximum output current $\leq$ 30mA)

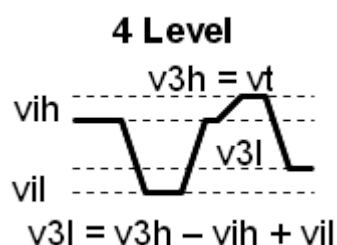
High level range	0.0 V to 2.5 V
Low level range	-0.8 V to 2.45 V
$v_t = v_{3h}$ voltage level range	-0.5 V to 2.5 V
Maximum level shift (v_{3h} voltage delta from V_{ih})	$\pm$ 850 mV
Swing range	50 mV to 1700 mV

DC Level accuracy @ 2 Level Mode

High voltage v_{ih}	$\pm$ 12 mV
Swing	$\pm$ 5 mV $\pm$ 1%
Differential Swing	$\pm$ 10 mV $\pm$ 1%

DC Level accuracy @ 3&4 Level Mode

High voltage v_{ih}	$\pm$ 20 mV if $v_{3h} \leq 2V$ $\pm$ 20 mV $\pm$ 0.1 $\cdot$ ($v_{3h} - 2V$) if $v_{3h} > 2V$
Swing	$\pm$ 13 mV $\pm$ 1%
Differential Swing	$\pm$ 18 mV $\pm$ 1%
Termination voltage $v_t =$ Shifted High voltage v_{3h}	$\pm$ 30 mV if $v_{3h} \leq 2V$ $\pm$ 30 mV $\pm$ 0.1 $\cdot$ ($v_{3h} - 2V$) if $v_{3h} > 2V$
Shifted Swing	$\pm$ 15 mV $\pm$ 3%
Shifted Differential Swing	$\pm$ 20 mV $\pm$ 3%



DC Level accuracy @ 4 Level Mode

^[12] measured for drive, no termination (v_{3h}) switch

PS9G Receiver

AC Performance

intrinsic transition time (20 to 80%) at 850 mV swing in 50 Ohm	60 ps $\pm$ 20 ps
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DC Performance Single Ended

Input Voltage and Threshold range	-0.5 V to 2.5 V
Threshold Voltage accuracy	± 7 mV
Threshold Voltage resolution	± 1 mV

DC Performance Differential

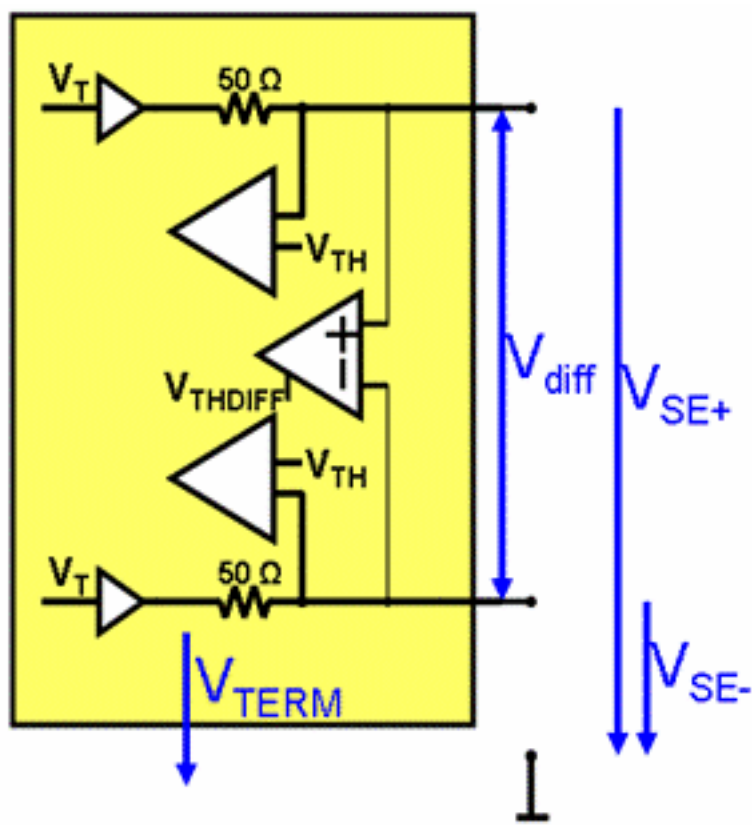
Input Voltage range	-0.5 V to 2.0 V
Threshold Voltage range ^[13]	-850 mV to +850 mV
Threshold Voltage accuracy for 0 mV	± 2 mV
Threshold Voltage accuracy ^[14]	± 5 mV $\pm 0.5\%$ of Threshold
Threshold Voltage resolution	± 1 mV

Termination

Single Ended / Center Tap Differential impedance	50 Ohm $\pm$ 2.5 Ohm
Termination level range	-0.1 V to 2.5 V
Termination Voltage accuracy	± 7 mV
Termination Voltage resolution	± 1 mV

^[13] Differential Threshold compares against the difference of VSE+ - VSE-

^[14] with common mode voltage $V_{CM} = (VSE+ + VSE-)/2$ in range 0..1V. If V_{CM} outside 0..1V, Threshold Voltage accuracy ± 7 mV $\pm 1\%$ of Threshold Voltage



PS9G Parametric Measurement Per-Pin PMU

PS9G Voltage force / measure

Voltage force / measure ranges

Range	-5.0 V to 8.0 V
Maximum input voltage range	-6.0 V to 9.0 V

Voltage force

Resolution	250 μ V
Accuracy in range -5.0 V to 8.0 V	$\pm(5 \text{ mV} + I \cdot R)$ ^[15]
Accuracy in range -0.0 V to 5.0 V	$\pm(2 \text{ mV} + I \cdot R)$

Compare mode

Resolution	500 μ V
Accuracy in range -5.0 V to 8.0 V	$\pm(2.5 \text{ mV} + I \cdot R) \pm 0.5\%$ of reading

Value measurement mode

Resolution	250 μ V
Relative accuracy in range 0.0 V to 5.0 V ^[16]	$\pm 3 \text{ mV}$
Accuracy in range 0.0 V to 5.0 V	$\pm(2 \text{ mV} + I \cdot R)$
Accuracy in range -5.0 V to 8.0 V	$\pm(5 \text{ mV} + I \cdot R)$

Voltage Clamp

Accuracy	$\pm(50 \text{ mV} + I \cdot R)$
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PS9G Current force / measure

	Range	Resolution	Measure accuracy	Force accuracy
Range 1 ^[17]	$\pm 60 \text{ mA}$	5 μ A	$\pm 90 \text{ } \mu\text{A} \pm 0.2\%$ of reading	$\pm 90 \text{ } \mu\text{A} \pm 0.15\%$ of setting
	$\pm 20 \text{ mA}$	5 μ A	$\pm 90 \text{ } \mu\text{A} \pm 0.1\%$ of reading	$\pm 90 \text{ } \mu\text{A} \pm 0.1\%$ of setting
Range 2	$\pm 2 \text{ mA}$	0.5 μ A	$\pm 2.0 \text{ } \mu\text{A} \pm 0.1\%$ of reading	$\pm 2.0 \text{ } \mu\text{A} \pm 0.1\%$ of setting

^[15] I is the actual current, R is the wiring resistance of $\leq 0.5 \text{ Ohm}$

^[16] In two consecutive voltage value measurements with different force currents, the accuracy of the calculated delta voltage

^[17] additional constraints apply for the sum of PPMU current of all channel; per board static(>1s):1A (=15.6mA per channel) dynamic (<1s):3A (=46.8mA per channel); per CC static(>1s):8A (=15.6mA per channel) dynamic (<1s):12A (=23.4mA per channel)

	Range	Resolution	Measure accuracy	Force accuracy
Range 3	±200 µA	5 nA	±280 nA ±0.1% of reading	±300 nA ±0.1% of setting
Range 4	±20 µA	1 nA	±60 nA ±0.1% of reading	±60 nA ±0.15% of setting
Range 5	±5 µA	250 pA	±30 nA ±0.2% of reading	±30 nA ±0.2% of setting

Current force for relative measurement

Relative accuracy^[18] ±30 µA ±0.3% of setting in range 1

Current Clamp

Accuracy 0 .. 15% of range

^[18] In two consecutive voltage value measurements with different force currents, the accuracy of the calculated delta of the forced currents

PS9G Time Measurement Unit Per-Pin TMU

DC performance

Specifications of receiver apply (see *PS9G Receiver* on page 17).

AC performance

Maximum input frequency	4 GHz
Linearity error (see <i>PS9G Linearity</i> on page 12)	±5 ps
Accuracy, single timestamp to ARM (t0)	±35 ps
Accuracy, skew/propagation delay (between different pin resources)	2x (t _{EPA} +35ps)
Intrinsic jitter (RMS)	5 ps
Resolution	1 ps
Other (Maximum number of simultaneously active TMUs: 128 shared within slots 1-4 and slots 5-8)	
Maximum Sample rate ^[19]	50 Mbps
Measurement Mode	Single ended & Differential

^[19] The maximum number of samples is limited by the available channel memory.

PS9G Simultaneous Bi-Directional SBD

PS9G SBD OTA/EPA

The tester drive swing is one third the swing of device output, speed up to 5Gbps.

Typical Per-Pin Pair OTA/EPA

Drive Receive Pin Pair Overall Timing Accuracy (t_{OTA})	± 40 ps
Drive Receive Pin Pair Edge Placement Accuracy ($t_{EPA}=t_{OTA}/2$)	± 20 ps

Typical Per Board OTA/EPA

Overall Timing Accuracy (t_{OTA})	± 70 ps
Edge Placement Accuracy ($t_{EPA}=t_{OTA}/2$)	± 35 ps

PS9G SBD Eye Height

Single Ended (with $V_{swing} = 500$ mV into 50 Ohm, tester drive swing is one third the swing of device output)

	2.5 Gbps	4 Gbps	5 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	85 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	80 % of V_{swing}	65 % of V_{swing}	45 % of V_{swing}

Differential (with $V_{swing} = 500$ mV into 50 Ohm, tester drive swing is one third the swing of device output)

	2.5 Gbps	4 Gbps	5 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	85 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	83 % of V_{swing}	68 % of V_{swing}	50 % of V_{swing}

PS9G SBD Driver

AC Performance

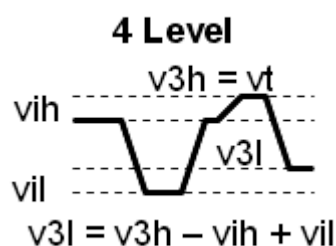
Transition time at 500 mV swing in 50 Ohm (20 to 80%) ^[20]	85 ps $\pm$ 20 ps
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DC Performance (specified into open and maximum output current ≤ 30 mA)

High level range	0.0 V to 1.8 V with termination > 0.2 V
	0.0 V to 1.5 V with termination ≤ 0.2 V

^[20] measured for drive, no termination (v3h) switch

Low level range	-0.2 V to 1.45 V with termination > 0.2 V 0.0 V to 1.45 V with termination ≤ 0.2 V
$v_t = v_{3h}$ voltage level range	0.0 V to 1.8 V
Maximum level shift (v_{3h} voltage delta from V_{ih})	±600 mV
Swing range	50 mV to 1200 mV
DC Level accuracy @ 2 Level Mode	
High voltage v_{ih}	±18 mV
Swing	±16 mV ± 5%
Differential Swing	±17 mV ± 4%
DC Level accuracy @ 3&4 Level Mode	
High voltage v_{ih}	±25 mV if $v_{3h} \leq 1.7$ V ±25 mV ± 0.1·($v_{3h} - 1.7$ V) if $v_{3h} > 1.7$ V
Swing	±15 mV ± 1%
Differential Swing	±20 mV ± 1%
Termination voltage v_t	±30 mV if $v_{3h} \leq 1.7$ V
= Shifted High voltage v_{3h}	±30 mV ± 0.1·($v_{3h} - 1.7$ V) if $v_{3h} > 1.7$ V
Shifted Swing	±20 mV ± 3%
Shifted Differential Swing	±25 mV ± 3%



PS9G SBD Receiver

AC Performance

intrinsic transition time (20 to 80%) at 850 mV swing in 50 Ohm	80 ps ± 20 ps
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DC Performance Single Ended

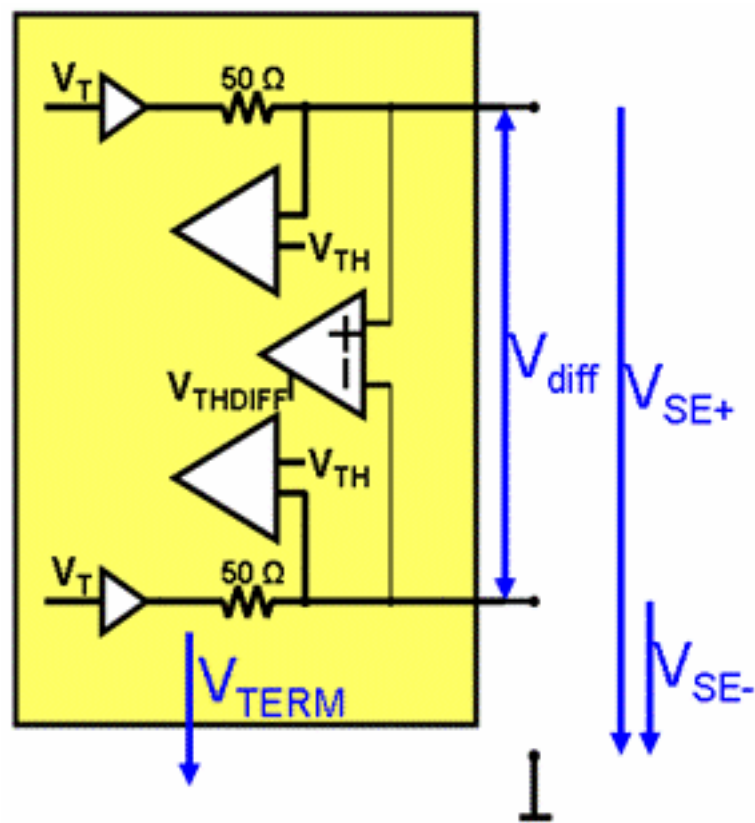
Input Voltage and Threshold range	0.0 V to 1.8 V
Threshold Voltage accuracy	±7 mV
Threshold Voltage resolution	±1 mV

DC Performance Differential

Input Voltage range	0.0 V to 1.8 V
Threshold Voltage range ^[21]	-850 mV to +850 mV
Threshold Voltage accuracy for 0 mV	±2 mV
Threshold Voltage accuracy ^[22]	±5 mV ±0.5% of Threshold
Threshold Voltage resolution	±1 mV

Termination

Single Ended / Center Tap Differential impedance	50 Ohm ± 2.5 Ohm
Termination level range	-0.1 V to 2.5 V
Termination Voltage accuracy	±7 mV
Termination Voltage resolution	±1 mV



^[21] Differential Threshold compares against the difference of VSE+ - VSE-
^[22] with common mode voltage VCM =(VSE+ + VSE-)/2 in range 0..1V. If VCM outside 0..1V, Threshold Voltage accuracy ±7 mV ±1% of Threshold Voltage